

Antimicrobial peptides are an important component of innate immunity. Since their discovery in 1939, over 5,000 peptides with antimicrobial properties have been isolated from organisms in every domain of life. They have demonstrated activity against enveloped viruses, bacteria, fungi, and even tumor cells and have recently gained interest for their potential in fighting antibiotic-resistant bacteria. Most known antimicrobial peptides are positively charged, but some with negative charges have also been discovered.

Frogs are known to secrete a variety of antimicrobial peptides that serve as a major source of immunity. Secretions from a new species of frog were collected and the peptides were separated by cation exchange chromatography. Four of the isolated peptides demonstrated strong antibacterial activity and were characterized by isoelectric focusing and sequencing:

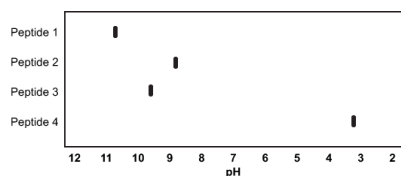


Figure 1 Results of isoelectric focusing of four antimicrobial peptides

Table 1 Amino Acid Compositions of Four Antimicrobial Peptides

Peptide	Sequence
Peptide 1	AALKGWTKSIPPKPSGKR
Peptide 2	FLPPSPTFKTSHKE
Peptide 3	IAKVALKAL
Peptide 4	ILGPVLGLVSDTLDDVLGIL

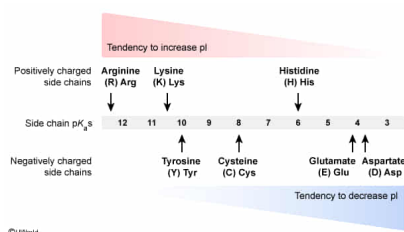
Which amino acid residue is most likely responsible for the decreased isoelectric point of Peptide 2 relative to Peptide 3?

- ☐ A. Threonine [7%]
- ☐ B. Histidine [21%]
- ☐ C. Aspartate [8%]
- ☒ D. Glutamate [63%]

Correct

62% answered correctly

Explanation:



A peptide's **isoelectric point (pI)** is the pH at which it has no net charge and therefore **does not migrate in an electric field**. This point is determined by the **ionizable amino acid side chains** present in the peptide. Seven side chains (R, K, H, Y, C, E, and D) are "**ionizable**," meaning their charges can be altered as protons are added or removed with varying pH:

- Ionizable side chains that contain oxygen or sulfur (Y, C, E, and D) are **negatively charged** when deprotonated and tend to **decrease pI**. For side chains that can be negatively charged, a lower pKa correlates with a stronger effect on pI. Therefore, **aspartate and glutamate** cause the greatest decreases in pI.
- Ionizable side chains that contain nitrogen (R, K, and H) are **positively charged** when protonated and tend to **increase pI**. Side chains that can be positively charged have a greater effect if the pKa is high, and therefore **arginine and lysine** cause the greatest increases in pI.
- All other side chains are nonionizable and do not affect pI.

Peptides 2 and 3 each contain two lysine (K) residues, but Peptide 2 also contains a glutamate (E) residue, which decreases its pI.

(Choice A) Threonine (T) does not have an ionizable side chain, so it does not affect a peptide's pI.

(Choice B) Histidine (H) can contribute only a positive charge to a peptide, so it can only increase pI.

(Choice C) Aspartate (D) can decrease pI, but it is not present in either Peptide 2 or Peptide 3.

Educational objective:

The isoelectric point is the pH at which a peptide is neutrally charged. Ionizable residues that can carry a negative charge, such as aspartate and glutamate, tend to decrease the pI. Those that can be positively charged, such as lysine and arginine, tend to increase it.

Time spent:0 sec

Subject:
Biochemistry

QID:400731

System:
Amino Acids and Proteins

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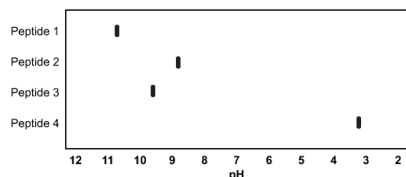


Figure 1 Results of isoelectric focusing of four antimicrobial peptides

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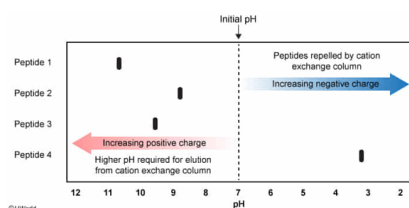
In which order will the four peptides elute from the cation exchange column as pH is increased from 7 to 12?

- ☐ A. Peptide 4, Peptide 3, Peptide 2, Peptide 1 [5%]
- ✓ ☒ B. Peptide 4, Peptide 2, Peptide 3, Peptide 1 [67%]
- ☐ C. Peptide 1, Peptide 2, Peptide 3, Peptide 4 [3%]
- ☐ D. Peptide 1, Peptide 3, Peptide 2, Peptide 4 [22%]

Correct

66% answered correctly

Explanation:



Ion exchange chromatography separates molecules by charge. The **beads** in **cation exchange** columns are **negatively charged** and **bind cations**, whereas anion exchange columns do the opposite. When a peptide has an isoelectric point (pI) that is higher than the pH of its surroundings, it is positively charged and will bind to a cation exchange column. As the pH increases beyond the pI, the peptide becomes negatively charged and will elute.

Figure 1 shows the pI values of each peptide. Peptide 4 has a pI that is lower than the initial pH of 7, so it is negatively charged and does not bind to the column. Rather, it is actively pushed out, or "sped on," as the negatively charged beads repel it; therefore, Peptide 4 will be the first peptide to elute (ie, exit from) the column. The other peptides have pI values above pH 7 and initially bind the column. Peptide 2 has the lowest pI of the bound peptides and elutes at the lowest pH (~8.7). Peptide 3 elutes next, at approximately pH 9.5, and Peptide 1 elutes at approximately pH 10.8.

(Choice A) Peptide 2 has a lower pI than Peptide 3, so it should elute at a lower pH.

(Choice C) Peptide 4 would elute first and Peptide 1 would elute last.

(Choice D) The peptides would elute from an *anion* exchange column in this order, which is exactly the reverse of cation exchange. In this scenario, anion exchange would require an initial pH of approximately 11 followed by decreasing pH so that all peptides would bind initially and elute separately.

Educational objective:

Cation exchange columns are negatively charged and bind cations. Increasing the pH can release bound molecules, and peptides with higher pI values require a higher pH to elute than peptides with low pI values. Anion exchange columns are the opposite.

Time spent:0 sec

Subject:
Biochemistry

QID:400732

System:
Amino Acids and Proteins

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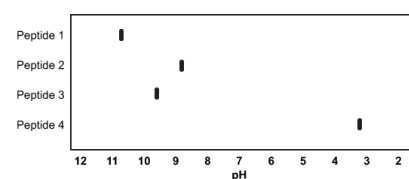


Figure 1 Results of isoelectric focusing of four antimicrobial peptides

Table 1 Amino Acid Compositions of Four Antimicrobial Peptides

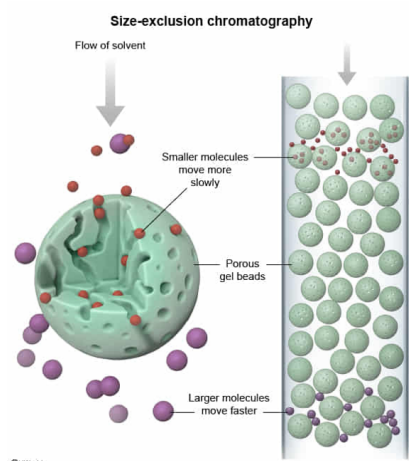
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Suppose the peptides had been purified by size exclusion chromatography instead of cation exchange. Which peptide would elute last?

- ☐ A. Peptide 1 [5%]
- ☐ B. Peptide 2
- ✓ ☒ C. Peptide 3 [71%]
- ☐ D. Peptide 4 [21%]

Correct
70% answered correctly

Explanation:



Like [gel electrophoresis](#), size exclusion chromatography separates molecules **by size**. Unlike gel electrophoresis, small molecules move more slowly through size exclusion columns than do large molecules.

The column is filled with tiny beads that have pores of varying sizes. Molecules that are too large to fit into any of the pores are

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"excluded" from some of the column volume and travel only through the space surrounding the beads, called the "**void volume**." Molecules that can fit into the large pores travel through the additional volume that they provide, and those that are small enough to fit inside all the pores travel through the entire available volume. Because they have a longer effective distance to travel before reaching the end of the column, **small molecules take longer to elute**. Peptide 3 is composed of only 9 amino acids and is substantially smaller than the other peptides, so it will elute last.

(Choices A, B, and D) These molecules are all much larger than Peptide 3 and will travel through less total volume. They will therefore elute sooner than Peptide 3.

Educational objective:

Size exclusion chromatography separates small molecules from large molecules. Smaller molecules take longer to elute because they effectively cross a greater distance than large molecules, which are excluded from the pores and travel only through the void volume.

Time spent:0 sec
Subject:
Biochemistry

QID:400733
System:
Amino Acids and Proteins

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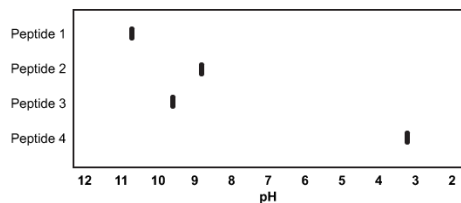


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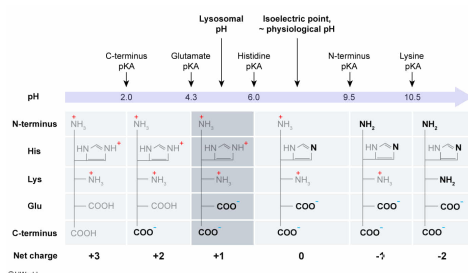
Antimicrobial peptides can be taken into cells by endocytosis. They are subsequently transported to the lysosome, where the pH decreases to approximately 5 and peptides are degraded. After lysosomal degradation, researchers found a tripeptide consisting of the last three amino acids of Peptide 2. What is the net charge of this tripeptide at lysosomal pH?

- ☐ A. -1 [8%]
- ☐ B. 0 [11%]
- ☒ C. +1 [64%]
- ☐ D. +2 [14%]

Correct

65% answered correctly

Explanation:



The last three amino acids of Peptide 2 are HKE. Each of these amino acids carries **ionizable side chains**, so they may contribute a charge, depending on the surrounding pH. The N- and C-termini of peptides are

also ionizable and may be charged. Determining the charge of each ionizable group requires that its **pKa** be compared with the surrounding pH, which is 5 in the lysosome. If the **pKa is higher than the pH**, the ionizable group is **protonated**. Protonation makes the net charge more positive, either by neutralizing a negative charge or by conferring a formal positive charge.

Histidine has a pKa of 6, so while it is deprotonated at cytosolic pH (7.4), it is **protonated at lysosomal pH** and carries a positive charge. Lysine and the N-terminus are also protonated and remain positively charged as the pH drops to lysosomal pH. Glutamate and the C-terminus have pKa values of ~4 and 2, respectively, so they remain deprotonated at lysosomal pH and are still negatively charged. The **sum of the charges** is therefore +1.

(Choice A) The negative charges of glutamate and the C-terminus are cancelled by positive charges from histidine, lysine, and the N-terminus, so the net charge cannot be negative.

(Choice B) HKE has a predominant net charge of 0 at cytosolic (physiological) pH, but not at lysosomal pH, because histidine becomes protonated in the lysosome.

(Choice D) The peptide would have a charge of +2 at lysosomal pH only if glutamate were absent.

Educational objective:

Ionizable side chains and peptide termini can be charged. If the pKa of an ionizable group is higher than the surrounding pH, the group is protonated and therefore more positive (or less negative). The net charge of a peptide at a given pH is the sum of the charges of all the ionizable groups.

Time spent: 0 sec

Subject:
Biochemistry

QID: 400734

System:
Amino Acids and Proteins

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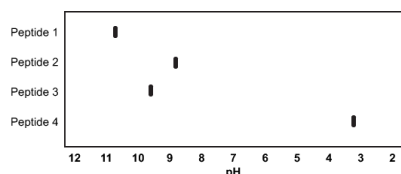


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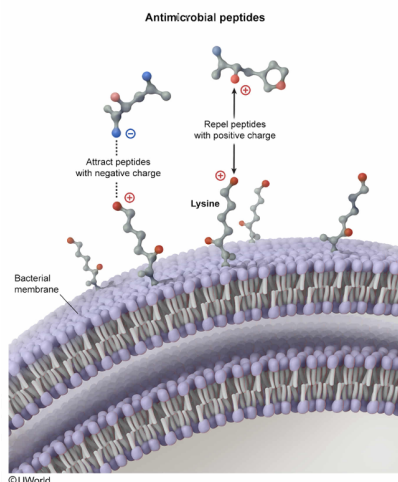
Some strains of *Staphylococcus aureus* modify their membranes by attaching lysine to phospholipids on the extracellular side. Given this information, which of the peptides tested is likely to be most effective against these strains?

- ☐ A. Peptide 1 [24%]
- ☐ B. Peptide 2 [7%]
- ☐ C. Peptide 3 [5%]
- ☒ D. Peptide 4 [61%]

Correct

62% answered correctly

Explanation:



As stated in the passage, most known antimicrobial peptides are positively charged. **Lysine** is also **positively charged** under **physiological conditions**, so bacteria coated with lysine can **repel** those antimicrobial peptides and therefore resist their activity. Negatively charged peptides, on the other hand, would be **attracted** to the positive charges of the lysine residues and

therefore would interact strongly with the membrane. Of the peptides tested, Peptide 4 is the only one with a net **negative charge** at physiological pH, having several **aspartate residues** (D). It therefore has the best chance of disrupting lysine-coated membranes.

(Choices A, B, and C) All these peptides have net positive charges and would therefore be repelled by the membrane. Even Peptide 2, with a negatively charged glutamate (E) residue, has a net positive charge at physiological pH as it has two lysine (K) residues.

Educational objective:

Amino acids may have positively or negatively charged side groups. Because similar charges repel and opposite charges attract, membranes coated with a positively charged amino acid such as lysine repel positively charged molecules but interact strongly with negatively charged residues such as aspartate.

Time spent:0 sec

Subject:
Biochemistry

QID:400735

System:
Amino Acids and Proteins

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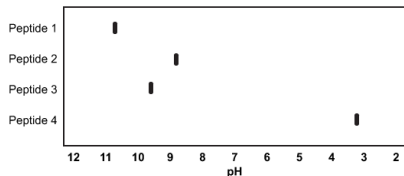


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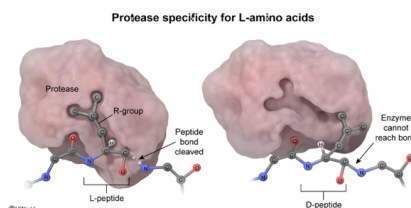
One of the challenges of antimicrobial peptides is their short physiological half-life. To solve this problem, some researchers have begun investigating synthetic antimicrobial peptides composed of D-amino acids. Why might these peptides have an increased half-life in the body relative to natural antimicrobial peptides?

- ✓ ☒ A. Proteases cannot act on peptides made of D-amino acids. [74%]
- ☐ B. D-amino acids are more soluble at physiological pH. [4%]
- ☐ C. Peptide bonds between D-amino acids require more energy to break. [18%]
- ☐ D. Hydrolysis of D-peptides is catalyzed by acids instead of bases. [1%]

Correct

74% answered correctly

Explanation:



With the exception of **glycine**, which has no chiral center, amino acids are **chiral molecules** that can exist in **two enantiomeric conformations**, designated **L and D forms**. In nature, amino acids are most frequently found in the L-conformation, so almost all **biomolecules** are designed to **work with L-amino acids**. Proteases typically recognize specific amino acid side chains and cleave peptide bonds near that side chain. When D-amino acids are used to make peptides, the side chain protrudes from the peptide at a different angle than it does in the L-conformation and is unrecognized by proteases. D-peptides therefore **remain uncleaved** and have a longer physiological half-life than their L counterparts.

(Choice B) D- and L-amino acids have essentially identical solubility.

(Choice C) A peptide bond is an amide bond between two amino acids. Amide bonds require approximately the same amount of energy to break, regardless of the stereochemistry of the chemical groups on either side of the bond.

(Choice D) Hydrolysis of both L- and D-peptides can be catalyzed by strong acids or strong bases. Because physiological pH is nearly neutral, these modes of catalysis are essentially absent under these conditions.

Educational objective:

Amino acids have chiral centers and can exist in two conformations, designated L and D. Almost all natural amino acids are synthesized in the L-conformation. As a result, enzymes essentially work exclusively with L-amino acids and do not affect D-peptides.

Time spent:0 sec

Subject:
Biochemistry

QID:400736

System:
Amino Acids and Proteins